CS:5980
Foundations of Embedded Systems

Dynamical Systems

Part I

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Dynamical Systems

Controller interacting with the physical world via sensors and actuators

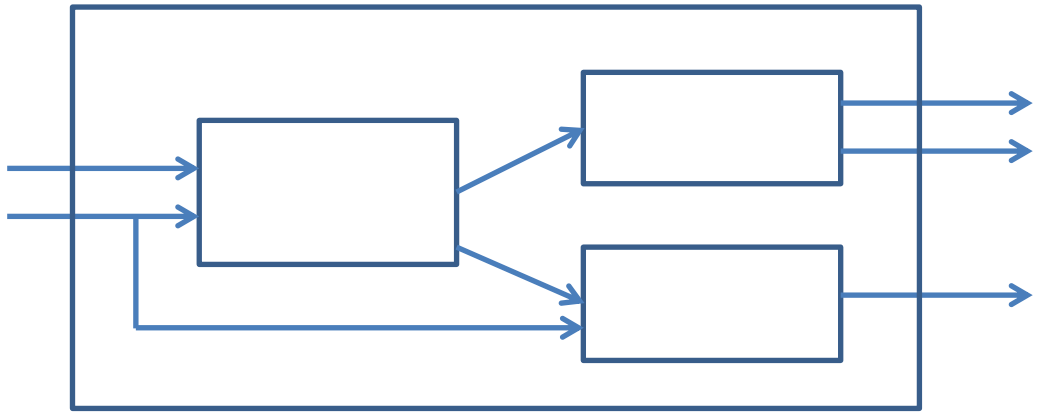
- Thermostat controlling temperature
- Cruise controller regulating speed of a car

System variables: (measures of) physical quantities evolving continuously over time

- Temperature, pressure, velocity …

Changes over time are modeled using differential equations

Model-Based Design



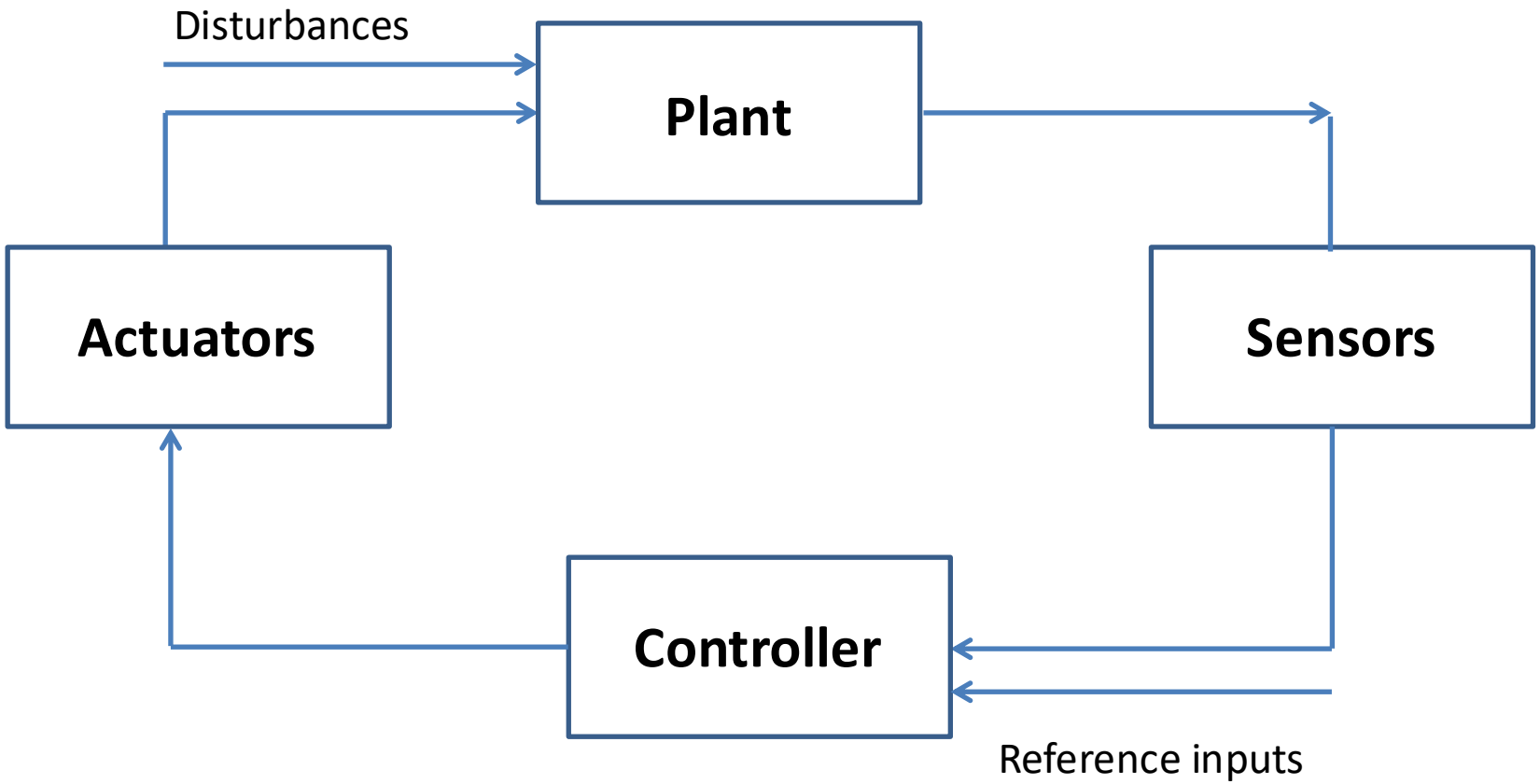
Block Diagrams

- Widely used in industrial design
- Tools: Simulink, Modelica, RationalRose…

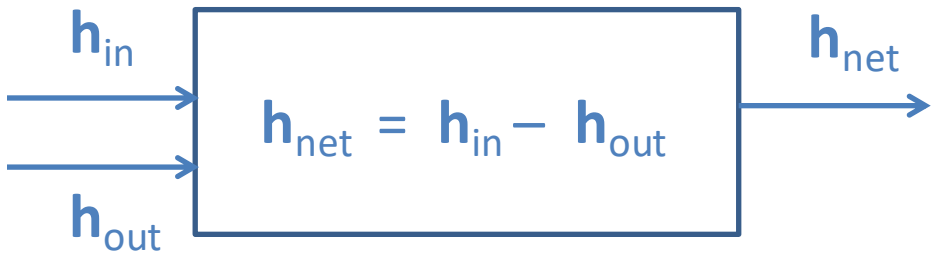
Key question: what is the execution semantics?

- Similar to synchronous model but continuous-time instead of discrete-time

Traditional Feedback Control Loop



Example: Heat Flow



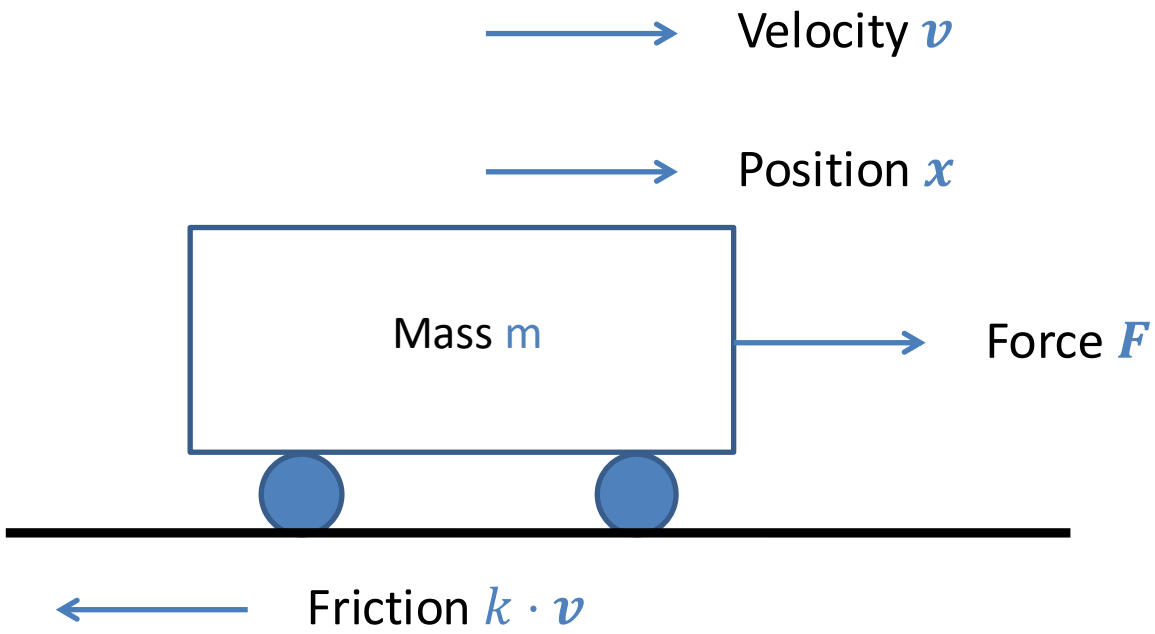
Input variables: h_{in} and h_{out} of type **real**

Output variable: h_{net} of type **real**

State variables: none

- *Signal*: assignment of values to variables as function of time t
- At each time t, value of output signal $h_{net}(t)$ equals $h_{in}(t) - h_{out}(t)$
- Output as a function of inputs/state, specified using algebraic equations (as opposed to assignments)

Car Model



- v, x and F are all functions of time t to space; k is a friction constant
- Newton's law of motion gives:

$$F - k \cdot dx/dt = m \cdot d^2x/dt$$

Notation

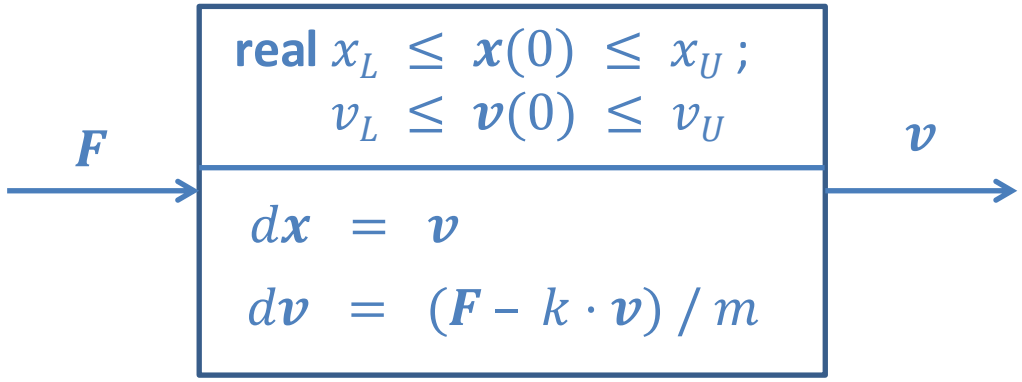
First derivative of function $f(t)$ with respect to t :

- $df(t)/dt$ (full notation)
- df/dt (with the understanding that f is a function of t)
- df (dimension t is implicit)
- f' (same as df)
- $\dot{f}$ (same as df)

Second derivative of function $f(t)$ with respect to t :

- $d^2f(t)/dt^2$ (full notation)
- d^2f/dt^2 (with the understanding that f is a function of t)
- d^2f (dimension t is implicit)
- f'' (same as d^2f)
- $\ddot{f}$ (same as d^2f)

Continuous-time Car Component



- The value of output variables is defined in terms of input and state variables
- For each state variable $\boldsymbol{s}$, its *rate of change* $d\boldsymbol{s}/dt$ is defined in terms of input and state variables

Executions of Car Component

Input signal: function $\boldsymbol{F}: \mathbb{R}_{\geq 0} \to \mathbb{R}$ that gives value of force over time

- should be continuous or piecewise-continuous

Given an initial state (x_0, v_0) and input signal $\boldsymbol{F}$, the execution of the system is defined by state-signals (i.e., functions)

$$\boldsymbol{x}: \mathbb{R}_{\geq 0} \to \mathbb{R} \quad \text{and} \quad \boldsymbol{v}: \mathbb{R}_{\geq 0} \to \mathbb{R}$$

that satisfy the *initial-value problem*:

$$\boldsymbol{x}(0) = x_0 \qquad d\boldsymbol{x}(t) = \boldsymbol{v}(t)$$
$$\boldsymbol{v}(0) = v_0 \qquad d\boldsymbol{v}(t) = d^2\boldsymbol{x}(t) = (\boldsymbol{F}(t) - k \cdot \boldsymbol{v}(t))/m$$

Executions of Car: Example 1

Suppose force $\boldsymbol{F}(t)$ is always 0, and initial position is 0.

We need to solve:

$$\begin{cases} \boldsymbol{x}(0) = 0 \\ \boldsymbol{v}(0) = v_0 \\ d\boldsymbol{x}/dt = \boldsymbol{v} \\ d\boldsymbol{v}/dt = k \cdot \boldsymbol{v}/m \end{cases}$$

Solution:

- Velocity decreases exponentially fast, converging to 0

$$\boldsymbol{v}(t) = v_0 \cdot e^{-k \cdot t/m}$$

- Position converges exponentially fast to $m \cdot v_0/k$

$$\boldsymbol{x}(t) = (m \cdot v_0/k)(1 - e^{-k \cdot t/m})$$

Executions of Car: Example 2

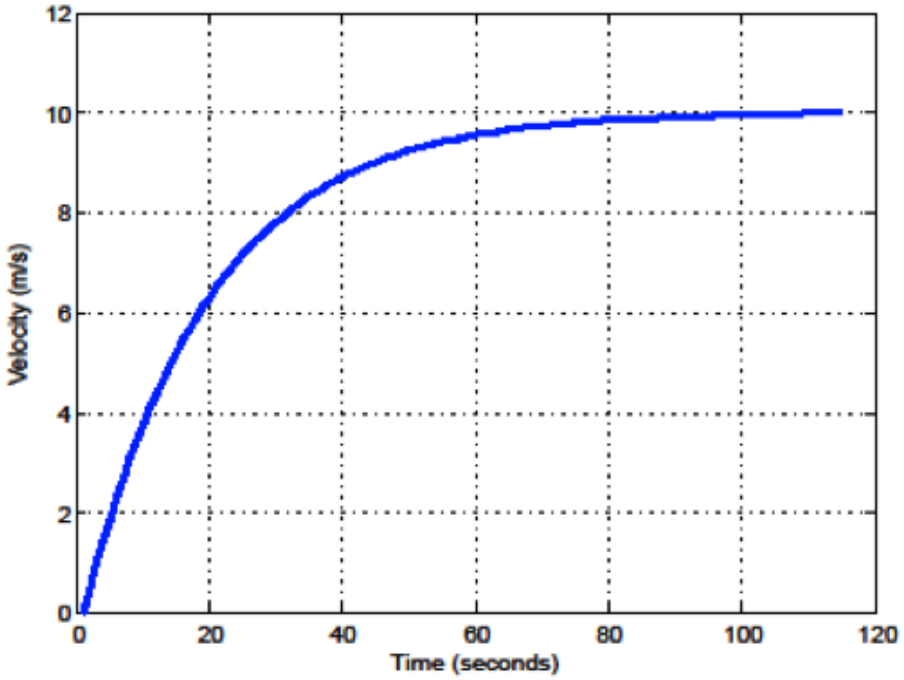
Suppose initial position is 0, initial velocity is 0, and force is constant F_0.

Then, to get executions, we need to solve:

- $\boldsymbol{x}(0) = 0$
- $\boldsymbol{v}(0) = 0$
- $d\boldsymbol{x}/dt = \boldsymbol{v}$
- $d\boldsymbol{v}/dt = (F_0 - k \cdot \boldsymbol{v})/mc$

Compute the solution using MATLAB

- Mass $m_c = 1000kg$
- Coefficient of friction $k = 50$
- Force $F_0 = 500\ Newton$
- Velocity converges to $10m/s$



Continuous-Time Component Definition

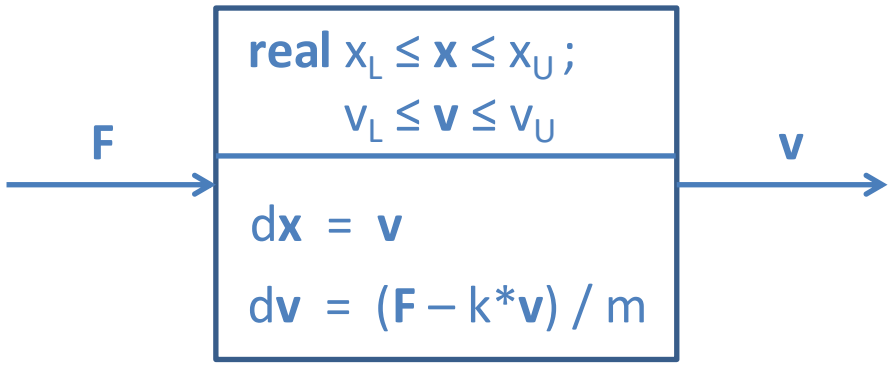
- Set I of real-valued *input variables*; type is either **real** (or real interval **real**[L, U])
- Set O of real-valued *output variables*
- Set S of real-valued *state variables*
- *Initialization* Init specifying set $[\![\text{Init}]\!]$ of initial states
- For each output var **y**, a real-valued expression h_y over I ∪ S
- For each state variable **x**, a real-valued expression f_x over I ∪ S

Execution

Given an input signal **I**(t) : $\textbf{real}_{>=0} \rightarrow \textbf{real}^{|I|}$, an *execution* consists of a differentiable state signal **S**(t) and output signal **O**(t) such that

1. **S**(0) is in $[\![\text{Init}]\!]$
2. For each output var **y** and time t, $\textbf{y}(t) = h_y(\textbf{I}(t), \textbf{S}(t))$
3. For each state var **x**, $d\textbf{x}(t) = f_x(\textbf{I}(t), \textbf{S}(t))$

Continuous-time Component Car



1. The value of output variables is defined in terms of input and state variables
2. For each state variable **s**, its *rate of change* ds is defined in terms of input and state variables

Existence and Uniqueness

Given an input signal **I**(t), when are we guaranteed the system has at least/exactly one execution?

- The input signal should be (at least piecewise) continuous but answer also depends on right-hand sides of equations defining state and output dynamics

This related to classical theory of Ordinary Differential Equations (ODEs)

Consider the initial value problem

$$d\mathbf{x}/dt = F(\mathbf{x}) \; ; \quad \mathbf{x}(0) = \mathbf{x}_0 \, , \qquad \mathbf{x}(t) \text{ is an } n\text{-dimensional vector}$$

- When do we have a unique differentiable function as a solution for **x**?

Solution Existence

***Initial value problem*:**

$$d\mathbf{x}/dt = F(\mathbf{x}) \;;\; \mathbf{x}(0) = \mathbf{x}_0 , \qquad \mathbf{x}(t) \text{ is } n\text{-dimensional vector}$$

The problem has a solution $\mathbf{x}(t)$ if function F is continuous

Example: of when solution does not exist:

$$dx/dt = \textbf{if } (x = 0) \textbf{ then } 1 \textbf{ else } 0$$

It is natural to require all right-hand-side expressions h_y and f_x in the definition of a continuous-time component to be continuous

- Will we see the discontinuous case in Chap. 9 (Hybrid Systems)

Continuous Function

Definition of continuity relies on a given notion of distance $\|\quad\|$ between points (e.g., Euclidean distance)

A function $f: \mathbb{R}^m \to \mathbb{R}^n$ is *(uniformly) continuous* if

for all $\varepsilon > 0$,

there is a $\delta > 0$ such that

for all $u, v \in \mathbb{R}^m$,

if $\|u - v\| < \delta$ then $\|f(u) - f(v)\| < \varepsilon$

Solution Uniqueness

Initial value problem:

$$d\mathbf{x}/dt = G(\mathbf{x}) \;;\; \mathbf{x}(0) = \mathbf{x}_0, \quad \mathbf{x}(t) \text{ is n-dimensional vector}$$

Theorem: There exists a unique solution $\mathbf{x}(t)$ if the function G is Lipschitz-continuous

Examples:

- A linear function such as $(\mathbf{F} - k\,\mathbf{v}) / m$ is Lip-continuous
- Quadratic function x^2 is Lip-continuous if domain of x is bounded

Counterexamples:

- $x^{1/3}$ is not Lipschitz-continuous: $dx/dt = x^{1/3}$; $x(0) = 0$ has multiple solutions:
  1. $\mathbf{x}(t) = 0$
  2. $\mathbf{x}(t) = (2t/3)^{3/2}$

Lipschitz-Continuous Function

Informally, a function f Lipschitz-continuous if
there is a constant upper bound on how much f's output changes

A function $f: \mathbb{R}^m \to \mathbb{R}^n$ is *Lipschitz-continuous* if there exists a constant c such that

for all $u, v \in \mathbb{R}^m$, $\quad \|f(u) - f(v)\| \leq c \cdot \|u - v\|$

Lipschitz-Continuous Component

Definition: A continuous-time component has *Lipschitz-continuous dynamics* if

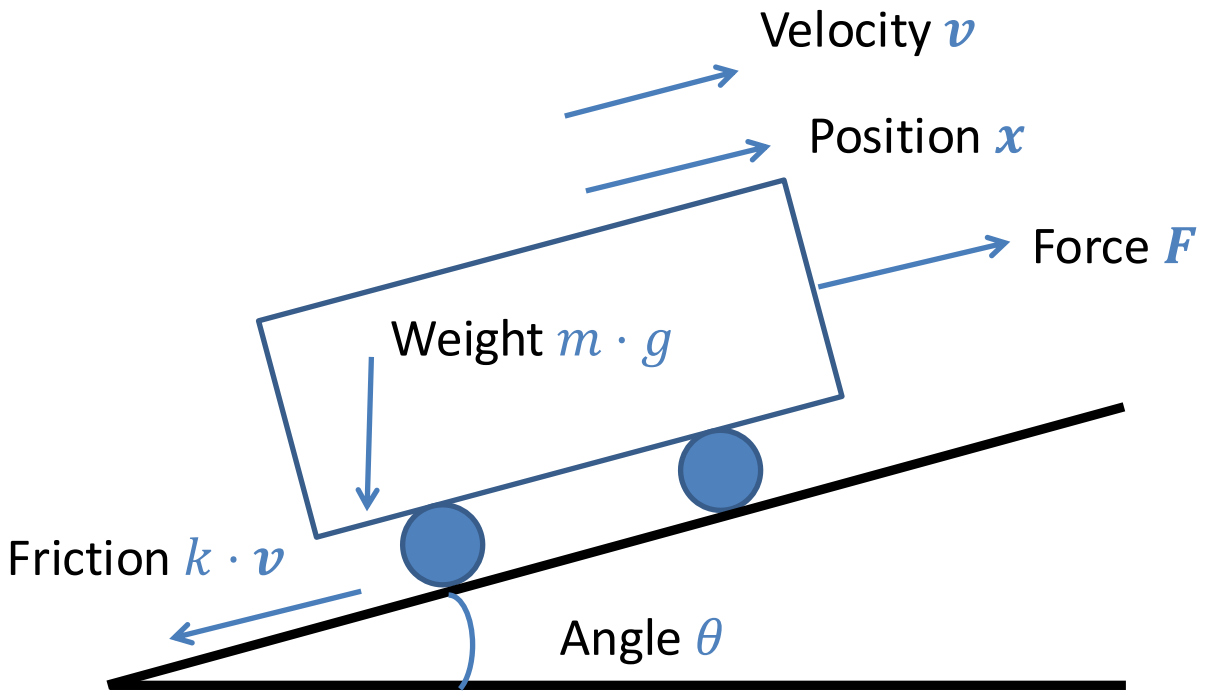
- each expression h_y corresponding to output variable **y** is a Lipschitz-continuous function of $I \cup S$
- each expression f_x corresponding to state variable **x** is a Lipschitz-continuous function over $I \cup S$

Theorem: Given a continuous input signal **I**(t), a component with Lipschitz-continuous dynamics has unique and continuous response signals **S**(t) and **O**(t)

Note: Continuity of output signals means they can be fed to other components in a block diagram

Henceforth, we will consider only Lipschitz-continuous components

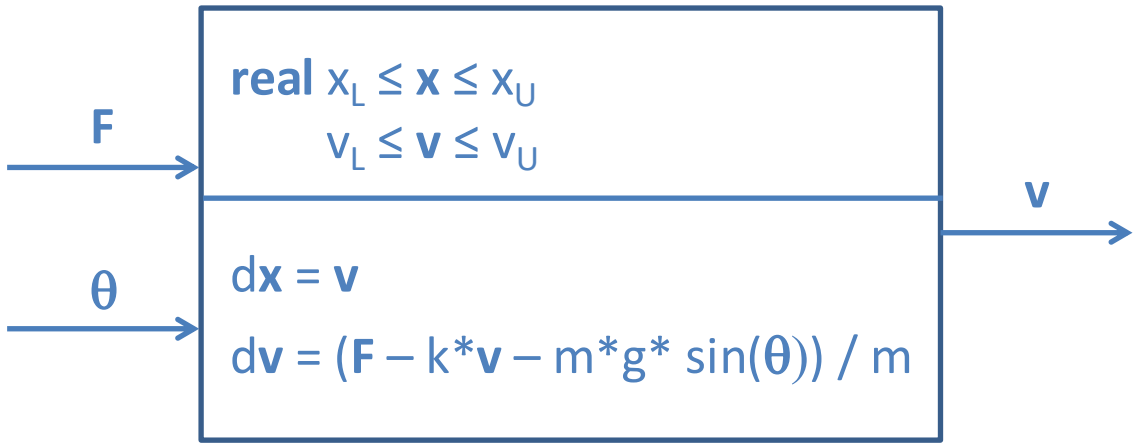
Car on a non-level road



Newton's law of motion gives

$$F - k \cdot v - m \cdot g \cdot \sin(\theta) = m \cdot d^2x/dt$$

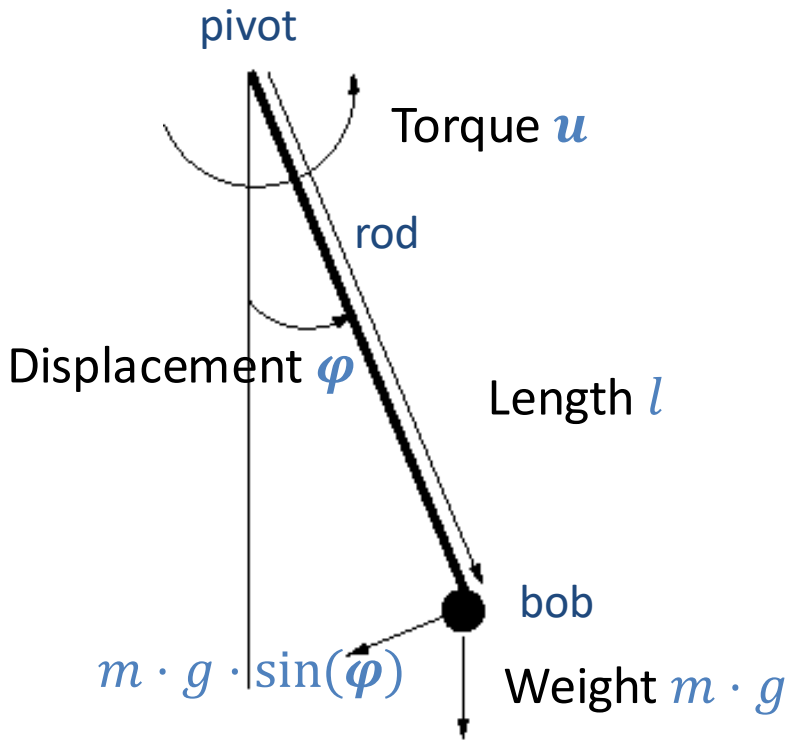
Continuous-time Car Component Revised



The road's slope, denoted by θ, models *disturbance*, or an uncontrolled input

Design problem: Find a controller with input **v** and output **F** such that the composed system works correctly for all continuous input signals $\theta(t)$, with $\theta(t)$ always in $[-\pi/6, \pi/6)$

Simple Pendulum



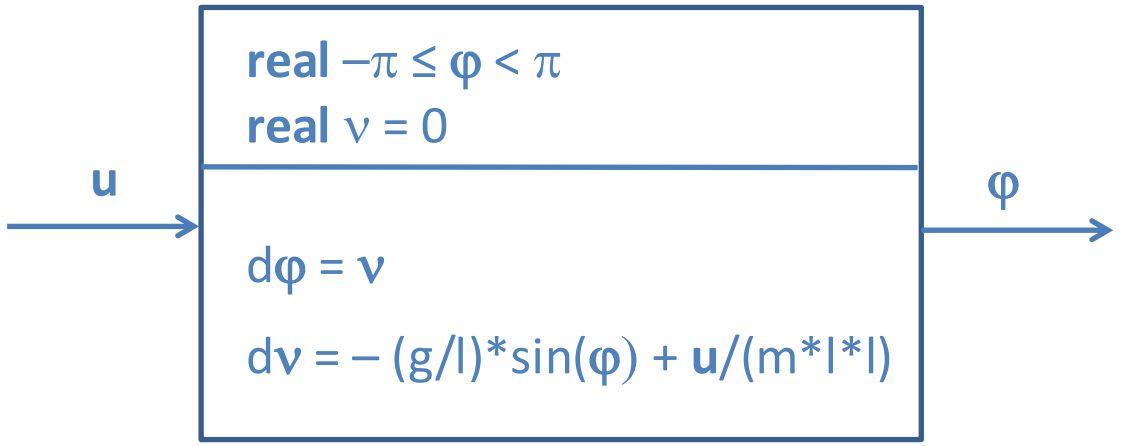
Assumptions: frictionless pivot, massless rod, massive bob

Dynamics captured by second-order non-linear differential equation:

$$m \cdot l^2 \cdot (d^2\varphi/dt^2) = u - m \cdot g \cdot l \cdot \sin(\varphi)$$

Pendulum Model

$$m \cdot l^2 \cdot (d^2\boldsymbol{\varphi}/dt^2) = \boldsymbol{u} - m \cdot g \cdot l \cdot \sin(\boldsymbol{\varphi})$$



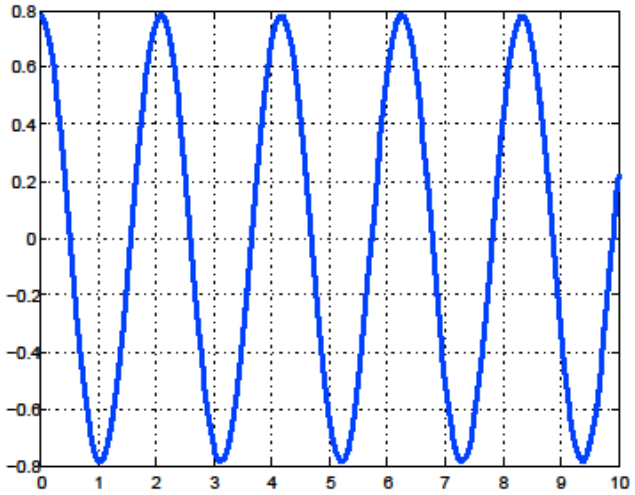
u: torque

φ: angular displacement (in radians, in $[-\pi, \pi)$)

ν: angular velocity

Angular Displacement

- External torque = 0
  Initial displacement = $\pi/4$
- Oscillatory motion plotted by MATLAB
- What are the equilibria of this pendulum?



Properties of Dynamical Systems

Correctness requirements for dynamical systems:

- Safety
- Liveness
- Stability

Example: Cruise controller

Safety: speed should always be within certain bounds

Liveness: actual speed should eventually converge to desired speed

Stability: as the road slope changes, speed should change gradually

Stability of Dynamical Systems

Intuitively, a dynamical system is *stable* if

small perturbations in the input cause only small changes in the output

Classical mathematical formalization of stability:

- Lyapunov stability of equilibria
- Bounded-Input-Bounded-Output stability of response

Stability is studied for *closed* continuous-time components, i.e., components with with no inputs

- If a component has inputs, we can analyze it by fixing them to a constant

Equilibria of Dynamical Systems

Consider a closed continuous-time component H

- Assume state **x** is n-dimensional, and its dynamics is Lipschitz-continuous and given by $d\mathbf{x}/dt = f(\mathbf{x})$

Definition: A state x_e is an *equilibrium* of H if $f(x_e) = 0$

Note: if a component H starts in an equilibrium state x_e, it stays in this state at all times

Pendulum Equilibria

Dynamics when $\mathbf{u}(t) = 0$

$$d\varphi = \nu$$

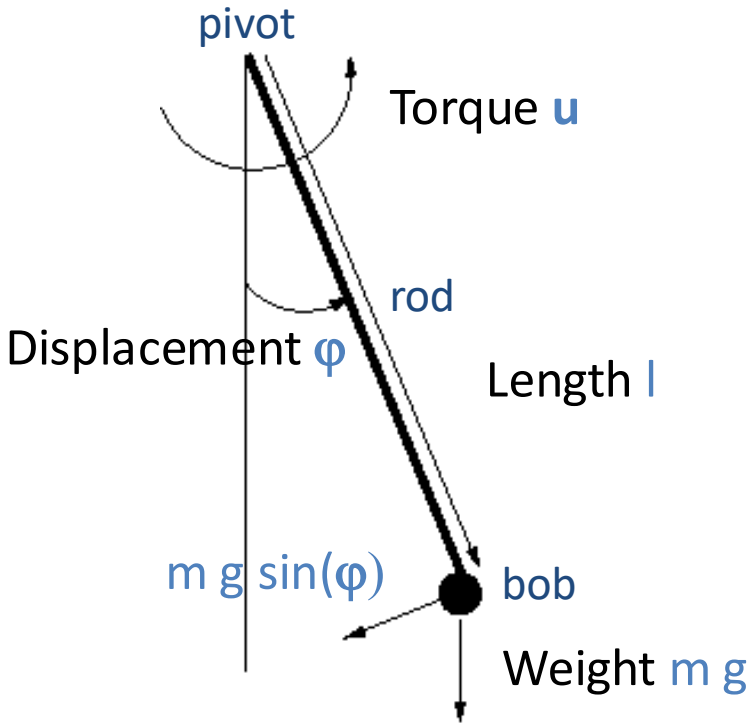
$$d\nu = -g \sin(\varphi) / l$$

Equilibrium states are solutions of:

$\nu = 0$

$\sin(\varphi) = 0$

$-\pi \le \varphi < \pi$



Equilibrium state 1: $\nu = 0, \varphi = 0$ Pendulum is vertically downwards

Equilibrium state 2: $\nu = 0, \varphi = -\pi$ Pendulum is vertically upwards

Lyapunov Stability

Consider a closed continuous-time component H with Lipschitz-continuous dynamics dx/dt = f(x)

Given an initial state s_0, let **x**[s] denote the *response signal*, the unique solution for the initial value problem

$$x(0) = s_0 \;;\; dx/dt = f(x)$$

Stability of an equilibrium: if the system is in an equilibrium state and we perturb its state slightly, as time passes,

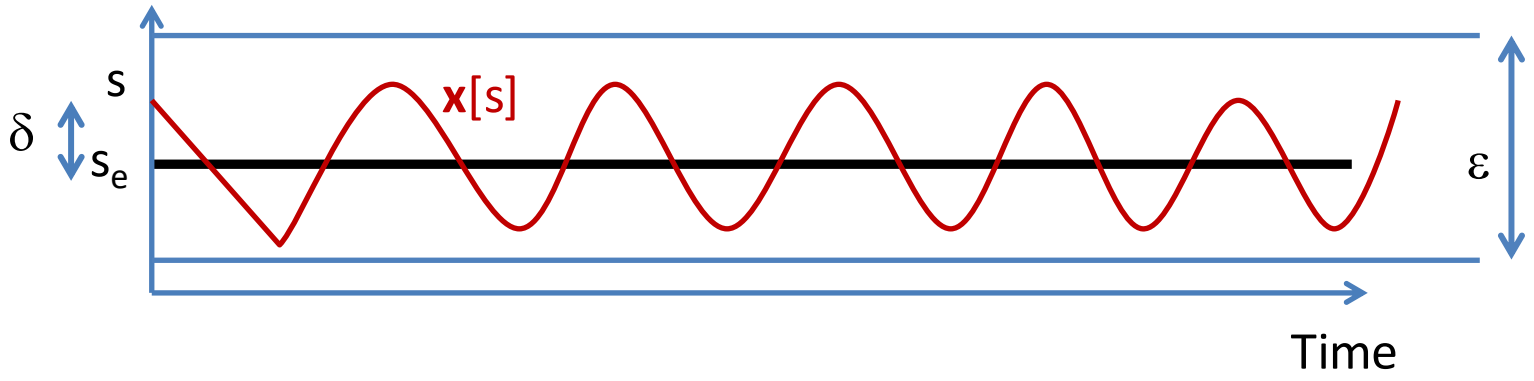
- will the system stay close to the equilibrium state ?
- will the system eventually return to that equilibrium state?

Lyapunov Stability Conditions

Note: if an initial state s_e is an equilibrium state then $\mathbf{x}[s_e](t) = s_e$ for all times t (i.e., it is a constant function)

Suppose another initial state s is close to s_e, do the states along the signal $\mathbf{x}[s]$ stay close to s_e as well? If so, s_e is said to be *stable*

Formally, s_e is *stable* if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all states s with $||s_e - s|| < \delta$ and times t, $||\mathbf{x}[s](t) - s_e|| < \varepsilon$

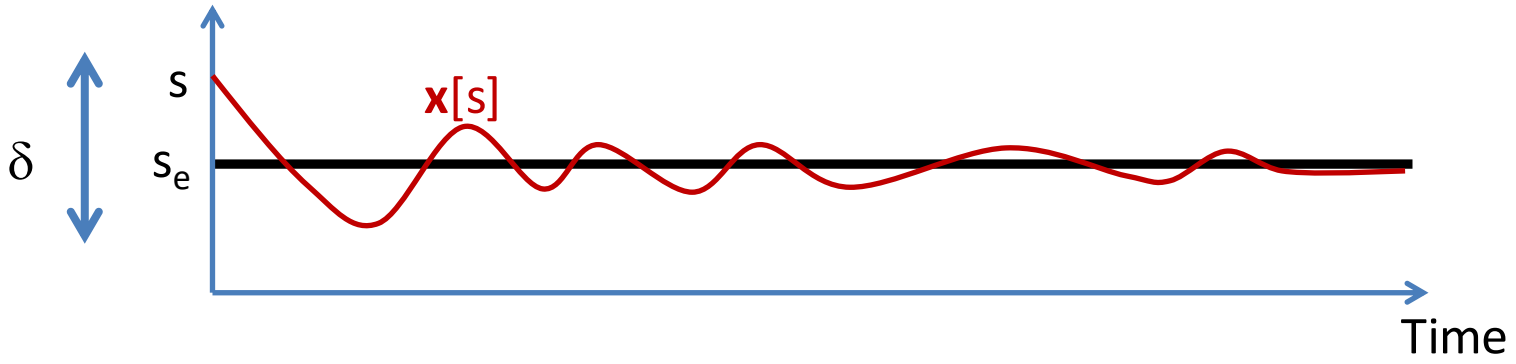


Lyapunov Stability Conditions

If, in addition, the response signal **x**[s] converges to the equilibrium state s_e, then s_e is *asymptotically stable*

Formally, s_e is *asymptotically stable* if

1. it is stable and
2. there exists a $\delta > 0$ such that for all states s with $||s_e - s|| < \delta$, $\lim_{t \to \infty} \mathbf{x}[s](t) = s_e$



Pendulum Equilibria

Equilibrium state 1: $\nu = 0$; $\varphi = 0$

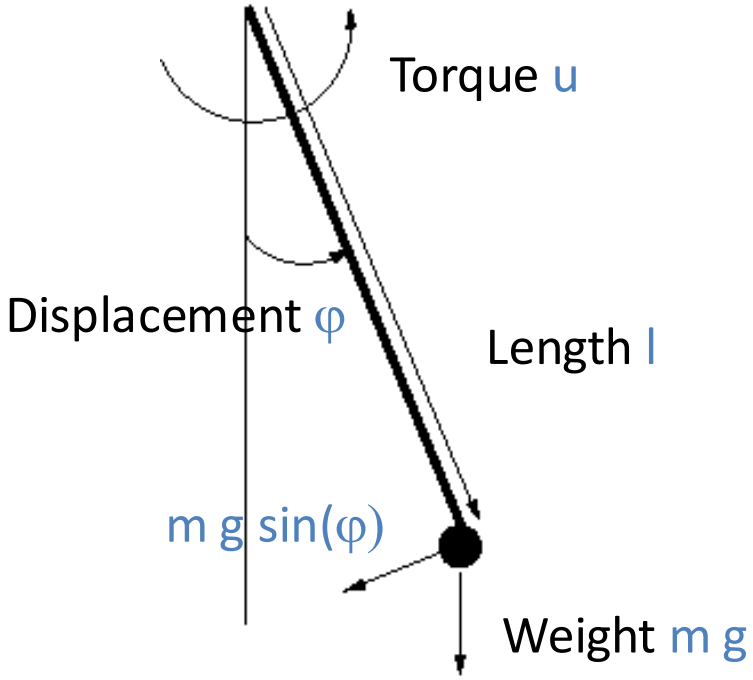
Pendulum is vertically downwards

Stable, but not asymptotically stable
(in frictionless model)

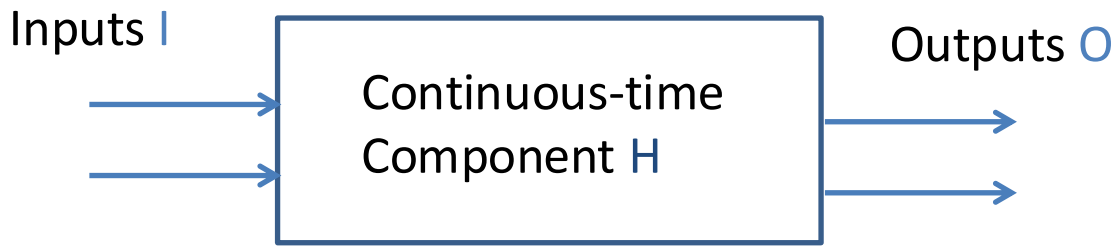
Equilibrium state 2: $\nu = 0$; $\varphi = -\pi$

Pendulum is vertically upwards

Unstable !



Input-Output Stability



A continuous-time component H maps input signals **I**(t) to output signals **O**(t)

Input-output stability: If we change the input signal slightly, the output signal should change only slightly

Suffices to focus on bounded signals

Input-Output Stability

Definition: A signal $\mathbf{x}(t)$ is *bounded* if there exists constant Δ such that

$||\mathbf{x}(t)|| \leq \Delta$ at all times t

Examples:

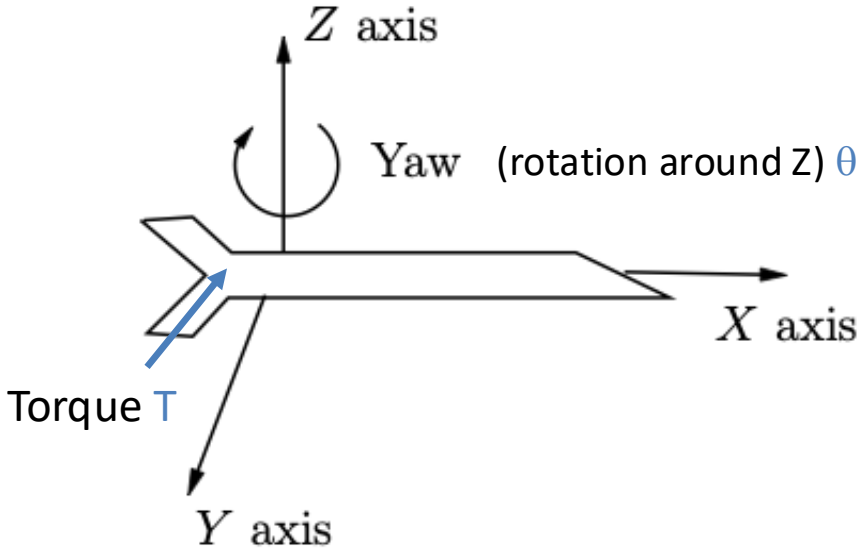
- Constant signal $\mathbf{x}(t) = a$: bounded
- Linearly increasing signal $\mathbf{x}(t) = a + bt$ with $b \neq 0$: not bounded
- Exponential signal $\mathbf{x}(t) = a + e^{bt}$ with $b \leq 0$: bounded
- Sinusoidal signals $\mathbf{x}(t) = a \sin(bt)$: bounded

A continuous-time component H with Lipschitz-continuous dynamics is *Bounded-Input-Bounded-Output (BIBO) stable* if

for every bounded input signal $\mathbf{I}(t)$, the output response signal $\mathbf{O}(t)$ from initial state $\mathbf{x}(0) = 0$ is bounded

Helicopter Model (Simplified)

Design problem: What torque should the tail rotor apply to keep helicopter from spinning?

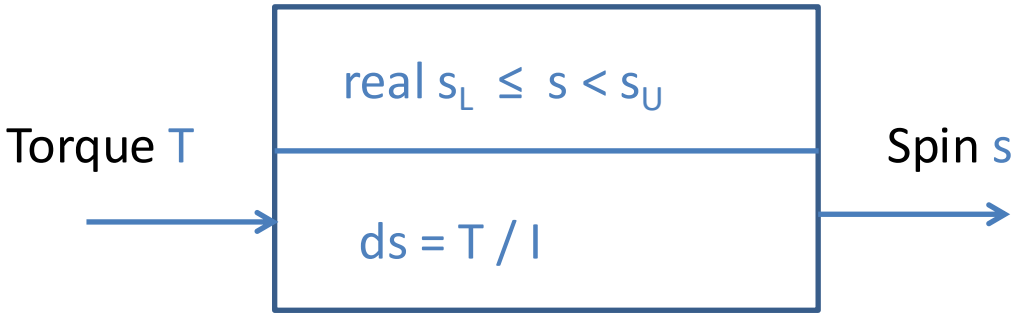


Spin ($d\theta$): s

Moment of inertia(*): I

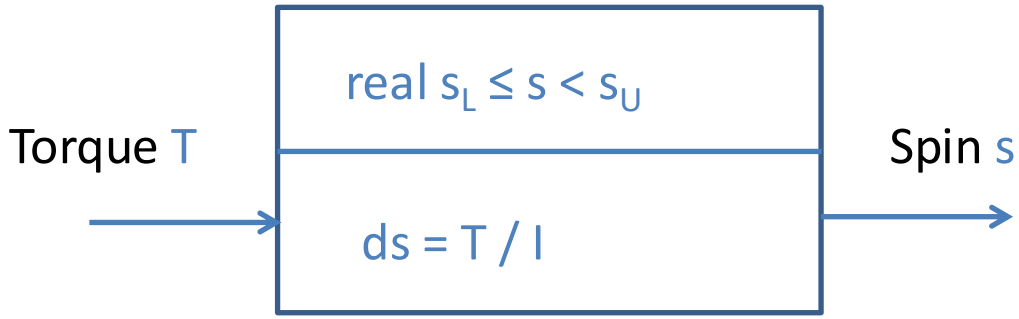
Equation of motion:

ds = T / I



(*) Constant for a rigid body

Stability of Helicopter Model



- Is the system BIBO stable?
- Consider bounded constant input signal **T**(t) = T_0
- Output response from initial state 0 not bounded: **s**(t) = (T_0 / I)t
- Not BIBO stable!
- What are the equilibria?
  - Set input torque to 0. If initial spin is c, it will stay c.
    Thus every initial state is an equilibrium state
  - Each such state c is stable but not asymptotically stable

Credits

Notes based on Chapter 6 of

Principles of Cyber-Physical Systems
by Rajeev Alur
MIT Press, 2015